MPI LANCHESTER–OSIPOV

Development plan, architecture, interface and indicator catalogue

Stages, timelines, technologies, project structure, data model, programme screens and the full catalogue of 253 indicators

Document 5 of 5 · Plan and catalogue

Data cut-off: October 2026 · Open sources
Military power assessment programme for states · project documentation

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1. Programme composition

The programme “MPI Lanchester–Osipov” consists of five modules. The first two implement the methodology (document 1), the third — coalitions (document 3), the fourth — flashpoint analysis (document 4), the fifth — data handling.

Module	Purpose	Result for the user
Military power index	253 indicators → 12 pillars → MPI with the nuclear axis, projection and intervals	rating, country profile, comparison, sensitivity to weights
Battle model	generalised Lanchester–Helmbold with reinforcements, Deitchman mixed law, salvo model, stochastic modelling	outcome probabilities, timelines, losses, required force ratio
Coalitions	assembling coalitions with reliability ρ, force share s, distance D and delay τ	coalition balance over time, alliance graph
Flashpoint monitor	tension index by country pairs, event feed	tension map, transition to the flashpoint scenario
Data	import, validation, reliability, versions, weapons and equipment reference, expert assessment log	coverage, data quality, reproducibility

2. Work plan

Stage	Timeline	Contents	Result and acceptance criterion
0. Specification	weeks 1–2	approve the pillars, roles, catalogue, formulas, expert score scales, the Admiralty scale for σ, alliance classes	methodology v1.0; catalogue frozen
1. Data	weeks 3–6	loading of open sources (World Bank, IMF, UN, SIPRI, EIA, UCS, UCDP, V-Dem), manual entry templates (IISS, expert assessments), weapons and equipment reference (300–500 models), treaty reference; pilot on 40 countries	panel 2015–2026, pilot coverage ≥ 85%
2. Index core	weeks 5–8	transformations, imputation, pillars with α, CES, deterrence, projection, Monte Carlo, AHP, entropy, PCA	formula property tests; reproducibility by hash
3. Battle model & coalitions	weeks 7–11	Helmbold with reinforcements, heterogeneous and salvo models, Deitchman, stochastic modelling, coalition module	agreement with the analytical solution of the square law to within 0.1%
4. MVP interface	weeks 6–12	porting the prototype to React, API, world map, coalition builder, data editor, AHP	MVP by week 10: rating, profile, comparison, battle model, coalitions for 40 countries
5. Calibration and validation	weeks 11–14	calibration of weights, α and ρ against wars and battles, validation against the wars of 2022–2026, comparison with CINC, GFP, DOE, MMP, expert survey	Brier and AUC no worse than DOE and MMP; Kendall's τ with experts ≥ 0.7
6. Flashpoint monitor	weeks 13–16	tension index, event feed, integration with UCDP and ACLED	20 flashpoints from document 4 with monthly updates
7. Scale and release	weeks 15–20	200+ countries, time series, DOCX/XLSX/PDF reports, roles and log, Windows build, documentation	version 1.0

Team: 1 methodologist-analyst, 1–2 Python developers, 1 front-end developer, 3–5 experts at stages 0, 5 and 6.

3. Architecture

3.1. Technologies

Layer	Technology	Why
Computational core	Python 3.12+: numpy, scipy (solve_ivp), pandas or polars, SALib	numerical methods, ODEs, Monte Carlo, sensitivity analysis
Storage	DuckDB (single user) or PostgreSQL (team)	analytical queries over the “country × indicator × year” panel
API	FastAPI	typed schemas, auto-generated documentation
Interface	React + TypeScript + Vite, ECharts, MapLibre GL, alliance graph (Cytoscape.js)	interactivity, fast recalculation
Import and export	openpyxl, python-docx	analysts working in Excel, reports in Word
Delivery	local web server or desktop application (Tauri or pywebview + PyInstaller)	operation in a closed environment
Tests	pytest + hypothesis, Playwright	reliability of calculations and interface

3.2. Project structure

```
lanchester-power/
├── docs/                    methodology, validation, Word documents
├── data/
│   ├── indicators.csv       indicator catalogue (source of truth), indicators.xlsx
│   ├── reference/           countries.csv, platforms.csv (weapons and equipment),
treaties.csv (alliances), anchors.csv
│   ├── raw/                 source extracts with a hash
│   └── processed/           panel.parquet: country × indicator × year × {value, σ,
flag}
├── core/                    the lanmil package
│   ├── etl/                 worldbank, imf, sipri, un, eia, ucdp, manual_xlsx
│   ├── quality.py, transform.py, impute.py
│   ├── weights/             ahp, entropy, pca, calibrate
│   ├── index.py             pillars, CES, deterrence, projection
│   ├── coalition.py         ρ, s, D, τ, contribution types
│   ├── lanchester/          laws, hetero, salvo, deitchman, stochastic, scenario
│   ├── hotspots.py          tension index
│   ├── uncertainty.py       Monte Carlo
│   └── validation/          backtest_wars, fit_battles, compare_indices
├── api/                     FastAPI
├── web/                     React application
├── prototype/               HTML prototype
└── tools/                   build_catalog.mjs, catalog_xlsx.py, word/
```

3.3. Data model

Table	Key fields
country	iso3, name*ru, name*en, region, bloc
indicator	id, pillar, subpillar, name, unit, dir, transform, role, weight, freq
source	id, name, url, licence, default_reliability
observation	country, indicator, year, value, low, high, source*id, reliability (A1–F6), is*imputed, note, entered*by, entered*at
platform / inventory	weapons and equipment model: class, generation, q*coef, method; inventory: country, platform, year, count, serviceable*share

Table	Key fields
treaty	id, name, parties[], class (A–F), article, signed, status, default_rho
alliance_event	treaty*id, date, type (participation, refusal, exercises), effect*on_rho, source
weight_profile / weight	weight profile (expert, entropy, calibrated, mixed) and values by node
scenario	side*a[], side*b[] with ρ, s, D, τ; theatre; d; α; horizon; break threshold; nuclear mode
hotspot	country pair, monthly tension index, events
run	id, type, parameters, methodology version, data hash, results

3.4. API methods

```
GET  /indicators?pillar=&role=           catalogue
GET  /countries/{iso3}/profile?year=     pillars, contribution, strengths and
weaknesses, battle model inputs
POST /index/compute                      {weights, rho, alpha, nuclear, norm, D,
mc_runs} → rating with intervals
POST /index/sensitivity                  Sobol indices by weight
GET  /treaties?country=                  the country's alliances with initial ρ
POST /coalition/compute                  {side, members[{iso3, rho, s, D, tau, types}]}
→ balance over time
POST /scenario/run                       {A, B, coalitions, conditions} → probabilities,
curves, invariants
GET  /hotspots                           tension index by flashpoint
POST /data/import                        Excel/CSV → validation → draft → publication
GET  /runs/{id}/report?format=docx|xlsx  report
```

4. Interface

4.1. Principles

1. **Explainability of every figure:** clicking a value shows the formula, its contribution to the total and the source with its reliability class.
2. **Uncertainty always in view:** the “whiskers” on the index, the rank range, the run band in the battle model.
3. **From general to particular:** rating → profile → comparison → coalitions → battle model → indicator.
4. **Scenario is the main mode:** any countries and coalitions are pitted against each other with one button.
5. **Instant feedback:** the weights, ρ, α and D sliders immediately recalculate the result and show rank shifts.
6. **Domain conventions:** side A in red, B in blue, as on operational maps; the terms CP, break threshold, force ratio.
7. Russian and English interface, light and dark themes, keyboard operation, responsiveness down to phone width.

4.2. Screens

Screen	Key elements	In the prototype
Rating	list with the MPI bar and 90% interval, rank range, 12-pillar heat strip, choice of measure and weights, distance D slider, world map	yes, without a map
Country profile	tiles (MPI, rank, excl. nuclear, λ, data), radar vs. median, pillars, strong and weak indicators, battle model inputs, indicator table with σ	yes
Comparison	2–5 countries: radars, advantage by pillar, largest gaps, transition to the battle model	yes, 2 countries
Coalitions	menu of ready-made coalitions, builder, ρ, s, τ sliders, alliance graph, balance over time, “what if the ally does not join”	planned
Battle model	sides and input editing, conditions, outcome probabilities, curves with a 10–90% band, invariant, sensitivity to α, equations with numbers	yes
Flashpoint monitor	tension map, event feed, history, “Open scenario” button	planned
Methodology & weights	weight sets, ρ, α, nuclear mode, normalisation, pillar weights, AHP with a consistency indicator, Sobol indices	yes, without AHP
Indicators	catalogue with search and filters, description of roles and sources	yes
Data	“country × indicator” coverage, Excel import, draft and publication, log, expert assessment form	planned
Reports	profile, comparison, coalition, scenario, flashpoint — in DOCX, XLSX, PDF	planned

4.3. User roles

- **Analyst** — ratings, comparisons, coalitions, scenarios, reports.
- **Expert** — expert scores and ρ with justification, participation in AHP and Delphi.
- **Data administrator** — import, validation, publication of snapshots, weapons and equipment and treaty references.
- **Manager** — summary screens, flashpoint monitor, reports.

5. Risks

Risk	Likelihood	Mitigation
Restricted access to quality and readiness data	high	expert assessments with explicit σ, data class in the interface
IISS Military Balance licence	medium	store derived estimates; manual entry with source attribution; cross-check against open data
Subjectivity of experts	high	AHP with consistency checking, Delphi, calibration against history, sensitivity analysis
False precision	medium	intervals everywhere; no single number without a range
Rapid obsolescence (wars change the balance within months)	high	quarterly update of “fast” indicators, a date for each value, monthly flashpoint monitor
Incorrect assessment of ally reliability	medium	alliance event log, “the ally does not join” scenarios
Political sensitivity and misuse	medium	transparent methodology, caveats in reports, version log

6. Indicator catalogue: summary

Code	Pillar	Indicators	Subpillars
GRD	Ground forces	32	Armour; Artillery and missiles; Anti-tank and army air defence; Army aviation and UAVs; Support and special forces; Readiness and quality
AIR	Air force & air defence	28	Combat aviation; Support aviation; Unmanned aviation; Armament; Readiness and infrastructure; Air and missile defence
NAV	Navy	25	Surface forces; Submarine forces; Naval aviation and coastal forces; Expeditionary capabilities; Quality and readiness
STR	Strategic forces	18	Nuclear arsenal; Delivery systems; Deterrence resilience; Strategic non-nuclear means
C4I	Space, cyber, command	20	Space; Cyber and information; Command and communications; Intelligence and EW
PER	Personnel	19	Numbers; Recruitment; Training and quality; Morale and resilience
ECO	Defence economy	21	Defence budget; Economic base; Financial resilience
IND	Defence industry & tech	24	Defence industry; Production capacity; Technology base; Raw materials
LOG	Logistics & mobilisation	23	Transport; Energy; Mobilisation; Civil resilience
GEO	Geography & demography	15	Territory; Position and frontiers; Demography
ALL	Alliances & presence	14	Alliances; Overseas presence; Partnerships and influence
GOV	Governance & experience	14	Combat experience; Governance; Military governance; Internal resilience
	Total	253	

Role	Meaning	Where it enters the model	Indicators
N	Numbers	N_0 (to the power of α)	41
Q	Quality	effectiveness e	57
M	Multiplier	effectiveness e (multiplicative)	35
S	Sustainment	reserve R, replacement r, delay	81
P	Projection	radius λ	13
G	Deterrence	deterrence axis, class 0–4	11
C	Context	terrain and defence coefficients	15

Notation in the full catalogue: “Dir.” — ↑ higher is better, ↓ higher is worse; “Transf.” — log (logarithm), lin (linear), share, score (expert score 0–10), cat (category); “Weight” — weight within the pillar 1–5.

> **Source of truth.** the catalogue is maintained in data/indicators.csv and data/indicators.xlsx. This document is a snapshot as of the build date; if the catalogue changes, rebuild the documents with node tools/word/build_all.js.

7. Full indicator catalogue

GRD — Ground forces (32)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
G01	Main battle tanks, total	Armour	units	↑	log	N	4	IISS
G02	Share of modern-generation tanks (3rd+ and upgraded)	Armour	%	↑	share	Q	3	IISS, Expert
G03	Average age of the tank fleet	Armour	years	↓	lin	Q	1	IISS, OSINT
G04	Infantry fighting vehicles (IFV)	Armour	units	↑	log	N	3	IISS
G05	Armoured personnel carriers, MRAPs and other armoured vehicles	Armour	units	↑	log	N	2	IISS
G06	Share of modern IFVs and APCs	Armour	%	↑	share	Q	2	IISS, Expert
G07	Tanks with active protection systems (APS)	Armour	%	↑	share	Q	1	OSINT, Expert
G08	Self-propelled artillery (SPH)	Artillery and missiles	units	↑	log	N	3	IISS
G09	Towed artillery	Artillery and missiles	units	↑	log	N	1	IISS
G10	Multiple launch rocket systems (MLRS)	Artillery and missiles	units	↑	log	N	3	IISS
G11	Heavy mortars (120 mm and above)	Artillery and missiles	units	↑	log	N	1	IISS
G12	Precision-guided weapons with range from 70 km (PGMLRS, tactical missile systems)	Artillery and missiles	units	↑	log	Q	4	IISS, OSINT
G13	Tactical missile launchers	Artillery and missiles	units	↑	log	N	3	IISS
G14	Counter-battery radars and artillery reconnaissance	Artillery and missiles	units	↑	log	M	2	IISS, Expert
G15	Anti-tank guided missile systems (ATGM)	Anti-tank and army air defence	units	↑	log	N	2	IISS
G16	Army short- and medium-range SAM systems	Anti-tank and army air defence	units	↑	log	N	3	IISS
G17	Man-portable air-defence systems (MANPADS)	Anti-tank and army air defence	units	↑	log	N	1	IISS, OSINT
G18	Counter-UAV means (EW, close-in weapon systems)	Anti-tank and army air defence	score	↑	score	M	2	Expert
G19	Attack helicopters	Army aviation and UAVs	units	↑	log	N	3	IISS
G20	Multi-role and transport helicopters	Army aviation and UAVs	units	↑	log	N	2	IISS
G21	Tactical reconnaissance UAVs	Army aviation and UAVs	units	↑	log	M	2	IISS, OSINT
G22	Loitering munitions and FPV drones (annual output)	Army aviation and UAVs	thousand units	↑	log	N	3	OSINT, Expert
G23	Special operations forces	Support and special forces	thousand people	↑	log	N	2	IISS

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
G24	Marines and airborne troops	Support and special forces	thousand people	↑	log	N	2	IISS
G25	Engineering equipment (bridgelayers, mine-clearing)	Support and special forces	units	↑	log	S	1	IISS
G26	Tactical electronic warfare systems	Support and special forces	units	↑	log	M	2	OSINT, Expert
G27	Units at permanent readiness (brigade equivalents)	Readiness and quality	units	↑	log	N	3	IISS
G28	Personnel manning of units	Readiness and quality	%	↑	share	Q	2	Expert, OSINT
G29	Equipment serviceability rate	Readiness and quality	%	↑	share	Q	3	Expert, OSINT
G30	Field training intensity (exercise days per year)	Readiness and quality	days	↑	lin	Q	2	Expert, OSINT
G31	Thermal imagers and night sights	Readiness and quality	score	↑	score	Q	1	Expert
G32	Artillery ammunition stocks	Readiness and quality	million rounds	↑	log	S	3	Expert, OSINT

AIR — Air force & air defence (28)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
A01	Combat aircraft, total	Combat aviation	units	↑	log	N	4	IISS
A02	4th/4+ generation fighters	Combat aviation	units	↑	log	N	3	IISS
A03	5th-generation fighters	Combat aviation	units	↑	log	Q	4	IISS
A04	Attack aircraft and tactical bombers	Combat aviation	units	↑	log	N	2	IISS
A05	Share of stealth platforms	Combat aviation	%	↑	share	Q	2	IISS, Expert
A06	Airborne early warning and control (AEW&C) aircraft	Support aviation	units	↑	log	M	4	IISS
A07	Aerial refuelling tankers	Support aviation	units	↑	log	P	3	IISS
A08	Heavy military transport aircraft	Support aviation	units	↑	log	P	3	IISS
A09	Medium and light military transport aircraft	Support aviation	units	↑	log	S	1	IISS
A10	Electronic warfare and signals intelligence aircraft	Support aviation	units	↑	log	M	3	IISS
A11	Trainer/combat aircraft	Support aviation	units	↑	log	S	1	IISS
A12	MALE and HALE strike UAVs	Unmanned aviation	units	↑	log	N	3	IISS, OSINT
A13	Long-range loitering munitions (annual output)	Unmanned aviation	thousand units	↑	log	N	2	OSINT, Expert
A14	Long-range air-to-air missiles (class)	Armament	score	↑	score	Q	2	Expert

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
A15	Air-launched cruise missile stocks	Armament	units	↑	log	S	2	Expert
A16	Precision-guided bombs and guided bomb units	Armament	thousand units	↑	log	S	2	Expert
A17	Average annual pilot flying hours	Readiness and infrastructure	hours	↑	lin	Q	3	Expert, OSINT
A18	Aircraft fleet serviceability	Readiness and infrastructure	%	↑	share	Q	3	Expert
A19	Average age of the combat aircraft fleet	Readiness and infrastructure	years	↓	lin	Q	1	IISS
A20	Military airfields with runways of 2,400 m or more	Readiness and infrastructure	units	↑	log	S	2	OSINT
A21	Hardened aircraft shelters	Readiness and infrastructure	score	↑	score	Q	1	OSINT, Expert
A22	International and joint flying exercises	Readiness and infrastructure	per year	↑	log	Q	1	OSINT
A23	Long-range SAM battalions	Air and missile defence	units	↑	log	N	4	IISS
A24	Medium-range SAM battalions	Air and missile defence	units	↑	log	N	3	IISS
A25	Surface-to-air missile stocks	Air and missile defence	thousand units	↑	log	S	3	Expert
A26	Non-strategic missile defence (interception of short- and medium-range ballistic missiles)	Air and missile defence	score	↑	score	M	3	OSINT, Expert
A27	Territory coverage by an integrated air-defence system	Air and missile defence	%	↑	share	M	3	Expert
A28	Long-range and over-the-horizon radars	Air and missile defence	units	↑	log	M	2	OSINT

NAV — Navy (25)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
N01	Total displacement of the combat fleet	Surface forces	thousand tonnes	↑	log	N	4	IISS, OSINT
N02	Aircraft carriers	Surface forces	units	↑	lin	P	4	IISS
N03	Cruisers and destroyers	Surface forces	units	↑	log	N	4	IISS
N04	Frigates	Surface forces	units	↑	log	N	3	IISS
N05	Corvettes	Surface forces	units	↑	log	N	2	IISS
N06	Missile and patrol boats	Surface forces	units	↑	log	N	1	IISS
N07	Vertical launch system (VLS) cells, total	Surface forces	units	↑	log	N	4	IISS, OSINT
N08	Naval area air defence	Surface forces	score	↑	score	Q	3	Expert

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
N09	Anti-ship missiles (range, speed)	Surface forces	score	↑	score	Q	3	Expert
N10	Mine countermeasure vessels	Surface forces	units	↑	log	N	1	IISS
N11	Nuclear-powered attack submarines	Submarine forces	units	↑	log	N	4	IISS
N12	Non-nuclear submarines (incl. AIP)	Submarine forces	units	↑	log	N	3	IISS
N13	Unmanned combat underwater vehicles	Submarine forces	score	↑	score	Q	1	OSINT, Expert
N14	Maritime patrol aircraft (ASW)	Naval aviation and coastal forces	units	↑	log	M	2	IISS
N15	Carrier-based fighters	Naval aviation and coastal forces	units	↑	log	N	3	IISS
N16	Coastal defence missile systems	Naval aviation and coastal forces	units	↑	log	N	2	IISS
N17	Unmanned surface vessels (USVs)	Naval aviation and coastal forces	score	↑	score	N	2	OSINT, Expert
N18	Coast guard (displacement)	Naval aviation and coastal forces	thousand tonnes	↑	log	C	1	IISS
N19	Amphibious assault and helicopter carriers	Expeditionary capabilities	units	↑	log	P	3	IISS
N20	Tank landing and other landing ships	Expeditionary capabilities	units	↑	log	P	2	IISS
N21	Combat support ships and naval tankers	Expeditionary capabilities	units	↑	log	P	3	IISS
N22	Naval bases, including overseas	Expeditionary capabilities	units	↑	log	P	2	OSINT
N23	Average age of the fleet	Quality and readiness	years	↓	lin	Q	2	IISS, OSINT
N24	Days at sea per ship per year	Quality and readiness	days	↑	lin	Q	2	Expert, OSINT
N25	Operational availability rate of ships	Quality and readiness	%	↑	share	Q	2	Expert

STR — Strategic forces (18)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
S01	Nuclear warheads, total	Nuclear arsenal	units	↑	log	G	4	SIPRI-YB, FAS
S02	Deployed strategic warheads	Nuclear arsenal	units	↑	log	G	4	SIPRI-YB, FAS
S03	Non-strategic nuclear weapons	Nuclear arsenal	units	↑	log	G	2	FAS, BAS
S04	Fissile material stocks (HEU, plutonium)	Nuclear arsenal	tonnes	↑	log	G	1	IPFM

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
S05	Latent nuclear potential (enrichment, reprocessing)	Nuclear arsenal	score	↑	score	G	1	IAEA, Expert
S06	ICBM launchers	Delivery systems	units	↑	log	G	3	SIPRI-YB, FAS
S07	SLBM launchers	Delivery systems	units	↑	log	G	3	SIPRI-YB, FAS
S08	Strategic bombers	Delivery systems	units	↑	log	G	2	IISS, FAS
S09	Nuclear triad (0–3 components)	Delivery systems	score	↑	cat	G	4	FAS, Expert
S10	Medium-range ballistic missiles	Delivery systems	units	↑	log	N	2	IISS, OSINT
S11	Assured retaliation (force survivability)	Deterrence resilience	score	↑	score	G	4	Expert
S12	Missile attack warning system	Deterrence resilience	score	↑	score	M	3	OSINT, Expert
S13	Strategic missile defence	Deterrence resilience	score	↑	score	M	2	OSINT, Expert
S14	Extended deterrence (allied "nuclear umbrella")	Deterrence resilience	score	↑	cat	G	2	Expert
S15	Hypersonic systems	Strategic non-nuclear means	score	↑	score	Q	3	OSINT, Expert
S16	Long-range land- and sea-based cruise missiles	Strategic non-nuclear means	units	↑	log	N	3	IISS, Expert
S17	Anti-satellite weapons (ASAT)	Strategic non-nuclear means	score	↑	score	Q	2	SWF, Expert
S18	Non-nuclear precision weapons with range from 1,500 km	Strategic non-nuclear means	score	↑	score	Q	2	Expert

C4I — space, cyber, command (20)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
C01	Military and dual-use satellites	Space	units	↑	log	M	3	UCS
C02	Imagery intelligence satellites (optical and radar)	Space	units	↑	log	M	3	UCS
C03	Military communications satellites	Space	units	↑	log	M	2	UCS
C04	Own satellite navigation system (GNSS)	Space	score	↑	cat	M	3	Expert
C05	SIGINT and early-warning satellites	Space	units	↑	log	M	2	UCS
C06	Orbital launches per year	Space	units	↑	log	S	2	OSINT
C07	Own spaceports and launch capabilities	Space	score	↑	cat	S	1	OSINT
C08	Cyber command (existence, personnel)	Cyber and information	score	↑	score	M	2	OSINT, Expert
C09	Global Cybersecurity Index (ITU GCI)	Cyber and information	score	↑	lin	M	2	ITU

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
C10	Offensive cyber capability	Cyber and information	score	↑	score	M	2	Expert
C11	Military use of AI (programmes, adoption)	Cyber and information	score	↑	score	Q	2	Expert
C12	Quantum technologies and cryptography	Cyber and information	score	↑	score	Q	1	Expert
C13	Integrated automated command system (network-centricity)	Command and communications	score	↑	score	M	4	Expert
C14	Secure military communications at all levels	Command and communications	score	↑	score	M	3	Expert
C15	Joint (combined) exercises per year	Command and communications	units	↑	log	Q	2	OSINT
C16	Strategic reconnaissance UAVs and aircraft	Intelligence and EW	units	↑	log	M	2	IISS
C17	Strategic electronic warfare systems	Intelligence and EW	score	↑	score	M	3	Expert
C18	Radar coverage of airspace	Intelligence and EW	%	↑	share	M	2	Expert
C19	Scale of military intelligence (resource estimate)	Intelligence and EW	score	↑	score	M	2	Expert
C20	Control of undersea cables and communications nodes	Intelligence and EW	score	↑	score	C	1	OSINT, Expert

PER — Personnel (19)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
P01	Active-duty military personnel	Numbers	thousand people	↑	log	N	4	IISS, WB
P02	Trained reserve	Numbers	thousand people	↑	log	S	3	IISS
P03	Paramilitary forces	Numbers	thousand people	↑	log	N	1	IISS
P04	Military personnel per 1,000 inhabitants	Numbers	people	↑	lin	N	1	WB, IISS
P05	Share of contract personnel	Recruitment	%	↑	share	Q	2	Expert, OSINT
P06	Conscription service term	Recruitment	months	↑	lin	C	1	CIA
P07	Military pay relative to average wage	Recruitment	times	↑	lin	Q	1	OSINT
P08	Personnel retention (share renewing contracts)	Recruitment	%	↑	share	Q	1	Expert
P09	Officer graduates from military academies per year	Recruitment	thousand people	↑	log	S	1	OSINT
P10	Training expenditure per service member	Training and quality	thousand $	↑	log	Q	3	SIPRI-M, Expert

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
P11	Share of professional NCOs	Training and quality	%	↑	share	Q	2	Expert
P12	Educational level of the officer corps	Training and quality	score	↑	score	Q	1	Expert
P13	Human Capital Index (World Bank)	Training and quality	score	↑	lin	Q	2	WB
P14	Share of personnel with combat experience	Training and quality	%	↑	share	Q	3	Expert
P15	Morale and unit cohesion	Morale and resilience	score	↑	score	Q	3	Expert
P16	Desertion and draft evasion	Morale and resilience	score	↓	score	Q	2	Expert, OSINT
P17	Medical fitness of the conscription pool	Morale and resilience	%	↑	share	S	1	OSINT
P18	Share of personnel costs in the defence budget	Morale and resilience	%	↓	share	C	1	SIPRI-M, NATO
P19	Foreign instructors and training abroad	Morale and resilience	score	↑	score	Q	1	Expert

ECO — Defence economy (21)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
E01	Military expenditure (current $)	Defence budget	billion $	↑	log	S	5	SIPRI-M
E02	Military expenditure by PPP	Defence budget	billion $ PPP	↑	log	S	3	SIPRI-M, IMF
E03	Military expenditure, % of GDP	Defence budget	%	↑	lin	S	2	SIPRI-M
E04	Military expenditure per capita	Defence budget	$	↑	log	Q	1	SIPRI-M
E05	Share of procurement and R&D in the defence budget	Defence budget	%	↑	share	Q	3	NATO, SIPRI-M, Expert
E06	Military R&D	Defence budget	billion $	↑	log	Q	3	OECD, Expert
E07	Five-year average annual growth of the defence budget	Defence budget	%	↑	lin	S	2	SIPRI-M
E08	Share of military expenditure in the state budget	Defence budget	%	↑	lin	S	1	SIPRI-M
E09	Cumulative military expenditure over 10 years	Defence budget	billion $	↑	log	S	3	SIPRI-M
E10	GDP (nominal)	Economic base	billion $	↑	log	S	3	IMF, WB
E11	GDP by PPP	Economic base	billion $ PPP	↑	log	S	3	IMF, WB
E12	GDP per capita by PPP	Economic base	$	↑	log	Q	2	IMF, WB
E13	Net resources (GDP × GDP per capita, Beckley)	Economic base	index	↑	log	S	3	IMF, Derived
E14	Five-year average annual GDP growth	Economic base	%	↑	lin	S	1	IMF

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
E15	Share of industry in GDP	Economic base	%	↑	share	S	2	WB
E16	Government debt	Financial resilience	% of GDP	↓	lin	S	2	IMF
E17	Foreign exchange reserves	Financial resilience	billion $	↑	log	S	2	IMF, WB
E18	External debt	Financial resilience	% of GDP	↓	lin	S	1	WB
E19	Inflation	Financial resilience	%	↓	lin	S	1	IMF
E20	State budget balance	Financial resilience	% of GDP	↑	lin	S	1	IMF
E21	Sanctions resilience and access to financial markets	Financial resilience	score	↑	score	S	2	Expert

IND — Defence industry & tech (24)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
I01	Revenue of national defence companies (SIPRI Top-100)	Defence industry	billion $	↑	log	S	3	SIPRI-I
I02	Arms exports (TIV, five-year total)	Defence industry	million TIV	↑	log	Q	3	SIPRI-AT
I03	Arms imports (TIV over five years) as dependence	Defence industry	million TIV	↓	log	S	2	SIPRI-AT
I04	Share of domestically produced weapons and equipment	Defence industry	%	↑	share	S	4	Expert, SIPRI-AT
I05	Full-cycle capability: weapon classes produced domestically	Defence industry	units	↑	lin	S	3	Expert
I06	Armoured vehicle output per year	Production capacity	units	↑	log	S	3	OSINT, Expert
I07	Artillery shell output per year	Production capacity	million units	↑	log	S	4	OSINT, Expert
I08	Combat aircraft output per year	Production capacity	units	↑	log	S	3	OSINT, Expert
I09	Missile output (cruise, ballistic, SAM) per year	Production capacity	units	↑	log	S	4	OSINT, Expert
I10	UAV output per year	Production capacity	thousand units	↑	log	S	3	OSINT, Expert
I11	Military shipbuilding (tonnage launched per year)	Production capacity	thousand tonnes	↑	log	S	3	OSINT
I12	Industrial mobilisation readiness	Production capacity	score	↑	score	S	3	Expert
I13	R&D expenditure	Technology base	% of GDP	↑	lin	Q	2	UNESCO, OECD
I14	Resident patent applications per million inhabitants	Technology base	units	↑	log	Q	1	WIPO
I15	Global Innovation Index	Technology base	score	↑	lin	Q	2	WIPO

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
I16	Graduates in engineering and STEM	Technology base	thousand people	↑	log	S	2	UNESCO
I17	Microelectronics: domestic chip production	Technology base	score	↑	score	S	3	Expert
I18	Domestic aero-engine industry	Technology base	score	↑	cat	S	2	Expert
I19	Machine tools and mechanical engineering (output)	Technology base	billion $	↑	log	S	2	UNIDO
I20	High-technology exports	Technology base	billion $	↑	log	Q	1	WB
I21	Steel production	Raw materials	million tonnes	↑	log	S	2	WSA
I22	Aluminium and titanium production	Raw materials	thousand tonnes	↑	log	S	1	USGS
I23	Rare earth metals (mining and processing)	Raw materials	tonnes	↑	log	S	2	USGS
I24	Propellant and explosive production capacity	Raw materials	score	↑	score	S	3	Expert

LOG — Logistics & mobilisation (23)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
L01	Railways	Transport	thousand km	↑	log	S	2	CIA, WB
L02	Paved roads	Transport	thousand km	↑	log	S	1	CIA
L03	Airports with paved runways	Transport	units	↑	log	S	1	CIA
L04	Major seaports	Transport	units	↑	log	P	1	UNCTAD, CIA
L05	Merchant fleet (deadweight)	Transport	million tonnes	↑	log	P	2	UNCTAD
L06	Inland waterways	Transport	thousand km	↑	log	S	1	CIA
L07	Military transport rolling stock (carriers, tractors)	Transport	score	↑	score	S	2	Expert
L08	Oil production	Energy	million bbl/day	↑	log	S	2	EIA
L09	Oil consumption	Energy	million bbl/day	↓	log	S	1	EIA
L10	Proven oil reserves	Energy	billion bbl	↑	log	S	1	EIA
L11	Natural gas production	Energy	billion m^3	↑	log	S	1	EIA
L12	Electricity generation	Energy	TWh	↑	log	S	2	EIA, IEA
L13	Primary energy consumption (CINC component)	Energy	EJ	↑	log	S	2	EIA, COW

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
L14	Energy self-sufficiency (production to consumption)	Energy	%	↑	lin	S	3	EIA, IEA
L15	Strategic oil reserve (days of imports)	Energy	days	↑	lin	S	1	IEA
L16	Mobilisation resource (fit men aged 18–49)	Mobilisation	million people	↑	log	S	3	UN, CIA
L17	Mobilisation reserve deployment time	Mobilisation	days	↓	log	S	3	Expert
L18	Equipment at storage bases	Mobilisation	thousand units	↑	log	S	2	IISS, OSINT
L19	Mobilisation stocks of materiel	Mobilisation	score	↑	score	S	2	Expert
L20	Food self-sufficiency	Civil resilience	%	↑	lin	S	2	FAO
L21	Power grid resilience (redundancy, protection)	Civil resilience	score	↑	score	S	2	Expert
L22	Civil defence and protective structures	Civil resilience	score	↑	score	S	1	Expert
L23	Hospital beds per 1,000 inhabitants	Civil resilience	units	↑	lin	S	1	WB, WHO

GEO — Geography & demography (15)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
D01	Territory area	Territory	thousand km²	↑	log	C	2	CIA, WB
D02	Strategic depth (border to key centres)	Territory	km	↑	log	C	3	GIS, Expert
D03	Defensive terrain (mountains, forests, water barriers)	Territory	score	↑	score	C	2	GIS
D04	Climatic severity for an attacker	Territory	score	↑	score	C	1	GIS
D05	Length of land borders	Position and frontiers	km	↓	log	C	1	CIA
D06	Coastline	Position and frontiers	km	↑	log	C	1	CIA
D07	Neighbours with territorial disputes	Position and frontiers	units	↓	lin	C	2	Expert, UCDP
D08	Access to the open ocean	Position and frontiers	score	↑	cat	P	2	GIS
D09	Control of straits and key sea lines of communication	Position and frontiers	score	↑	score	P	2	Expert
D10	Population	Demography	million people	↑	log	S	4	UN
D11	Annual number reaching conscription age	Demography	thousand people	↑	log	S	3	UN, CIA
D12	Urban population (CINC component)	Demography	million people	↑	log	S	1	UN, COW
D13	Median age	Demography	years	↓	lin	S	1	UN

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
D14	Total fertility rate	Demography	children/ woman	↑	lin	S	1	UN
D15	Dependency ratio	Demography	%	↓	lin	S	1	UN

ALL — Alliances & presence (14)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
X01	Membership in a bloc with a collective defence commitment	Alliances	score	↑	cat	M	4	ATOP, Expert
X02	Allies under mutual defence treaties	Alliances	units	↑	log	M	2	ATOP
X03	Combined military power of allies (reliability-weighted)	Alliances	index	↑	log	M	4	Derived
X04	Reliability of allied commitments	Alliances	score	↑	score	M	3	Expert
X05	Interoperability with allies (standards, exercises)	Alliances	score	↑	score	M	2	Expert
X06	Military bases and basing points abroad	Overseas presence	units	↑	log	P	3	OSINT
X07	Military personnel deployed abroad	Overseas presence	thousand people	↑	log	P	2	IISS
X08	Allied troops on own territory	Overseas presence	thousand people	↑	log	M	2	IISS
X09	Partners in military-technical cooperation	Partnerships and influence	units	↑	log	M	2	SIPRI-AT
X10	Recipients of arms exports	Partnerships and influence	units	↑	log	M	1	SIPRI-AT
X11	Intelligence sharing with allies	Partnerships and influence	score	↑	score	M	2	Expert
X12	Permanent membership of the UN Security Council	Partnerships and influence	score	↑	cat	C	1	UN
X13	Participation in UN peacekeeping operations	Partnerships and influence	people	↑	log	Q	1	UN
X14	Sanctions pressure (countries applying sanctions)	Partnerships and influence	units	↓	log	C	2	OSINT

GOV — Governance & experience (14)

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
H01	Interstate conflict experience over 15 years	Combat experience	score	↑	score	Q	3	COW, UCDP

ID	Indicator	Subpillar	Unit	Dir.	Transf.	Role	Weight	Sources
H02	Counter-insurgency and expeditionary experience	Combat experience	score	↑	score	Q	2	UCDP, Expert
H03	Battle casualty replacement rate	Combat experience	score	↑	score	S	2	Oryx, Expert
H04	Government effectiveness (WGI)	Governance	score	↑	lin	Q	2	WB
H05	Corruption Perceptions Index (CPI)	Governance	score	↑	lin	Q	2	TI
H06	Political stability and absence of violence (WGI)	Governance	score	↑	lin	S	2	WB
H07	Quality of military doctrine and its fit to modern war	Military governance	score	↑	score	Q	3	Expert
H08	Quality of senior command	Military governance	score	↑	score	Q	3	Expert
H09	Decision speed and delegation of authority	Military governance	score	↑	score	Q	2	Expert
H10	Transparency and accountability of the defence budget	Military governance	score	↑	score	C	1	TI-DS, SIPRI-M
H11	Internal armed conflicts (UCDP)	Internal resilience	score	↓	score	S	2	UCDP
H12	Public support for the army and defence	Internal resilience	score	↑	score	S	2	Polls, Expert
H13	Resilience to adversary information operations	Internal resilience	score	↑	score	S	1	Expert
H14	Risk of a military coup and politicisation of the army	Internal resilience	score	↓	score	C	1	V-Dem, Expert